



Data Science & Engineering
Analytics Technical Section (DSEATS)
Africa Region

DSEATS AFRICA REGION

Datathon 2026

For Industry Professionals, Young Professionals, and Students across Africa

REGISTRATION COMMENCES
20th of May, 2026

COMPETITION CLOSES
29th June, 2026

- ✓ All submissions must be entered by 29th June 2026 at 12:00 midnight GMT+1
- ✓ Full Competition guidelines, datasets and submission information will be sent to participants on 29th of May
- ✓ You are encouraged to form teams of 3 to 5 max.

Register here : <https://tinyurl.com/2026datathonreg>

GUIDELINES FOR PARTICIPATING TEAMS

(GENERAL & SUBMISSION GUIDELINES)

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1.0 General Guidelines

1.1 Overview

The SPE Data Science and Engineering Analytics Technical Section (DSEATS) Africa Datathon Challenge presents an exciting opportunity for participants to harness the power of machine learning to address real-world challenges in the oil and gas industry. Organized by the Society of Petroleum Engineers (SPE) Data Science and Engineering Analytics Technical Section (DSEAT) Africa, this dataset is designed to help geoscientists, engineers, data scientists, and energy analysts develop models to predict the existence of oil in unexplored areas based on geological and geophysical factors.

1.2 Challenge Objective

Participants are expected to work in teams of 3 to 5 persons (max) from at least 2 different organizations and/or schools to develop a machine learning (ML) model that accurately predicts the presence of oil in unexplored areas using certain geologic / seismic properties. The teams are expected to explore innovative approaches in this competition.

1.3 Key Components

- **About the Dataset:** Participants will have access to a dataset curated based on real-world geological principles and containing 5,000 (train + test) samples representing different locations around the world with varying geological and seismic properties. Each row in this dataset represents a specific location and below is a brief description of the features in the dataset:
 - **Rock_Type:** The primary rock formation in the area. It can be Sandstone, Limestone, or Shale. Sandstone and Limestone are generally better reservoirs, while Shale is less favorable.
 - **Porosity (%):** This tells us how much empty space exists in the rock. Higher porosity means more room to store oil. Values range from 0% to 30%.
 - **Permeability (mD):** Measures how easily fluids can move through the rock. Higher permeability means better oil flow, while low permeability makes extraction difficult. Ranges from 1 to 1000 millidarcies (mD).
 - **Trap_Type:** Traps help prevent oil from escaping or migrating upwards, making them crucial for accumulation in a reservoir. This dataset covers three (3) different types of structural traps: Anticline (folded rock layers), Fault (displaced rock layers) and Dome (circular rock uplift) None (no significant trap structure)
 - **Seismic_Score:** A score (0 to 1) based on seismic survey data, which detects underground structures. Higher scores indicate a higher likelihood of oil reservoirs.
 - **Proximity_to_Oil_Field (km):** The distance to the nearest known oil field. Closer proximity (0 to 5 km) increases the chances of finding oil, as geological conditions may be similar.
 - **Estimated_Reservoir_Depth (m):** The predicted depth (500 to 5000m) of the oil-bearing rock layer, estimated using seismic data. Deeper reservoirs are harder to extract but might hold more oil.

- **Oil_Presence:** The target variable (0 or 1). 1 = Oil Found and 0 = No Oil. This is based on a mix of geological rules and some random variation to simulate real-world uncertainty.

Note: This dataset contains outliers, null values, and some technical anomalies. It is expected that participating teams will apply oil and gas domain knowledge to clean and prepare the data for their machine learning workflows.

- **Evaluation Criteria:** Submissions will be evaluated based on the accuracy of the classification models, application of domain knowledge, the creativity and effectiveness of the approaches employed as well as the level of collaboration amongst team members.

1.4 How to Participate

Step 1: Register for the completion using this link: <https://tinyurl.com/2026datathonreg>

- Interested SPE members (Students and Professionals) can register for the challenge and will subsequently be provided with the datasets and competition guidelines. By clicking on the registration link above

Step 2: Read through this general and submission guidelines document.

- Please read the general guideline and the submission guideline carefully and be sure to understand and accept the conditions stated in the guidelines. Individuals who violate the guidelines will be automatically disqualified from the competition.

Step 3: Build appropriate ML model to analyze the dataset and provide two (2) classification results / predictions as follows:

- **Part-1:** Analyse the dataset provided following the standard machine learning workflow which should include (but not limited to): data preparation, exploratory data analysis, data preprocessing, feature engineering, modelling and classification. This part is purely data science based.
- **Part-2:** This phase mirrors Part 1 but requires a physics-informed approach to data preprocessing and feature engineering. Here, teams must leverage oil and gas domain expertise to detect and correct geological anomalies or inconsistencies in at least three (3) dataset features. These corrections must be applied to the training data prior to modeling and classification. Because this section heavily emphasizes domain knowledge and innovation, it will be a key focus area for the competition judges and teams are expected to demonstrate the value of applying domain knowledge to solving this problem.

Step 4: Submit your results as a team following the Submission Guidelines below. Towards the end of the selection process, the top teams being considered for the awards may be invited for an online interview.

1.5 Eligibility

- All members of Society of Petroleum Engineers (SPE) affiliated with any of the SPE Sections in Africa are eligible to participate.
- Past winners of SPE DSEATS Africa Region datathon may participate, provided that no two (2) past winners are on the same team.
- Close family and friends of SPE DSEATS Africa board members are not eligible to participate.

2.0 Submission Guidelines

1. Participating teams are expected to develop a machine learning model in **Python** programming language to accurately classify the dataset. The ML model should be developed using Google Colab or Jupyter Notebook (.ipynb) file format.
2. The Google Colab file should be accessible and submitted alongside with the oil presence classification (in csv format) and presentation slides summarizing the work done.
3. Submission shall be on a team basis - individual submissions will **NOT** be accepted. Registered participants are expected to collaborate and form teams of 3 to 5 persons (max) from at least 2 different organizations and/or schools. Each team should adopt a unique but decent, single-phrase team name having not more than 15 letters/characters.
4. Teams are expected to save their classification results / predictions in two(2) csv files using the following naming convention:
 - **TeamName_Prediction_Part1.csv** (for Part-1 result)
 - **TeamName_Prediction_Part2.csv** (for Part-2 result)
5. Teams should adopt the following naming conventions for both the PowerPoint slide and Google Colab / Jupyter Notebook file.
 - **TeamName_Presentation.pptx** (for PowerPoint slide deck)
 - **TeamName_PythonCode.ipynb** (for Google Colab or Jupyter Notebook file)

Note each team should submit only one (1) Google Colab or Jupyter Notebook file
6. Each team is expected to create a brief PowerPoint presentation of not more than **ten (10)** slides in total including the Title, Outline, Exploratory Data Analysis (EDA), Methodology, Results, Contributions and Thank You slides. A “Thank you” slide is not necessary and can be omitted. *(Note: Presentation slides that are more than 10 may result in deduction of points which will affect your team’s overall score)*
7. The **Title** slide of your team’s presentation should be captioned (e.g *Prediction of Oil Presence Using Machine Learning*), followed by your team’s name, names of team members and their respective SPE numbers, SPE Sections in Africa that they are affiliated to, School or Company affiliations and date. *Pls ensure that all these information are clearly captured on the title page otherwise some points shall be deducted.*
8. The **EDA** slide(s) should show relevant statistical plots
9. The **Methodology** slide(s) should show the ML workflow(s) used to achieve the classification results / predictions for both Part-1 and Part-2. Teams should provide summary of the data preprocessing and feature engineering done for both parts.
10. The **Results** slide(s) of your team’s presentation should amongst other things show summary of the evaluation metrics (ie accuracy) of the different models implemented.
11. The **Contributions** slide should **clearly** summarize the work done by each team member.
12. The PowerPoint presentation slide deck, the Google Colab file (or Jupyter Notebook file), and the Oil Presence prediction files in csv format should be sent to the email address below on or before the submission deadline: **Monday, 29 June 2026 by 11:59PM WAT.**
Email: speafricadseat@gmail.com *(Note that submissions after this deadline will not be accepted)*
13. The Subject of the email should read as follows:
TeamName_“SPE DSEATS Africa 2026 Datathon Submission”
e.g Innovisors_SPE DSEATS Africa 2026 Datathon Submission

The body of the email should be in simple English as shown below:

Dear Reviewer,

On behalf of team (name), find attached to this mail, presentation slide deck, the Google Colab file containing our solution, and the two (2) oil presence prediction.csv files.

Important Notes:

- a. Please, do not forget to attach the four (4) files (PowerPoint presentation slide deck, the Google Colab File, and the two (2) oil presence prediction.csv files)
- b. Only one member of the team should make the submission on behalf of the team and copy the other team members. Multiple submissions by different team members will attract penalties and possible disqualification.
- c. Only one final submission will be accepted, pls indicate which is your team's final one if you have made more than one submission. Not more than 2 submissions in all should be made over the competition period. Teams making more than 2 submissions will be disqualified.
- d. **FOLLOW ALL INSTRUCTIONS (BOTH THE GENERAL AND SUBMISSION GUIDELINES)! All the best!**

3.0 Datasets Provided

All registered participants in this competition shall be provided with two dataset files for training and testing as follows:

- 1) Training dataset: ***oil_presence_trainset.csv***
- 2) Test dataset: ***oil_presence_testset.csv***

Using the features in the training dataset, participating teams are expected to develop their best machine learning model and then use this to predict oil presence in the test dataset.

4.0 Evaluation Criteria

The Judges will assess the effectiveness of the machine learning techniques employed and the potential impact of the predictive models in the oil and gas industry. The following are the evaluation criteria that would be used.

- **Effectiveness of the ML Techniques:** Judges will evaluate the efficiency of the machine learning techniques / algorithms used.
- **Classification Accuracy:** The classification model's accuracy and performance would be evaluated with key performance indicators such as Accuracy, Precision, Recall, F1-Score etc
- **Innovation and Creativity:** Additional points shall be given to teams who adopt innovative and creative machine learning approaches and domain knowledge to solve this challenge. *However, please note that the use of Artificial Intelligence (AI)-assisted techniques or Generative AI is highly discouraged and will result in disqualification of any team involved. All submissions would first be curated using our AI detection tool.*
- **Team Participation & Presentation:** The level of teamwork, collaboration, clarity of presentation and specific contribution made by each team member will count towards a team's overall score.

5.0 Intellectual Property

Participants retain ownership of their submissions; however, by participating in this competition, participants grant the organizers the right to showcase and promote their work within the SPE community.

6.0 Rewards & Benefits

The top three (3) teams shall receive the following benefits:

- **Amazing Cash Prizes:** The competition promises to reward the top three (3) teams with amazing cash prizes.
- **SPE NAICE Awards:** In addition, the winning teams shall be recognised and awards presented to each team member during the Awards Night Ceremony at the forthcoming 2026 SPE NAICE Conference holding in Nigeria, at Eko Hotel & Suites Lagos from 3 to 5 August 2026.
- **Industry Recognition:** Outstanding submissions will be recognized and showcased within the SPE community, offering participants visibility and professional recognition in SPE Africa, and SPE International.
- **Professional Development:** Winners and participating teams will have the opportunity to enhance their machine learning skills, gain practical experience with application of machine learning techniques to solving typical oil and gas problem and network with industry experts.